United States Patent Application Publication	20250269761
Kind Code	A1
Publication Date	August 28, 2025
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INFORMATION PROVIDING SYSTEM

Abstract

An information providing system is a system that provides carbon footprint information of a battery pack mounted on a vehicle and includes one or more processors. The one or more processors are configured to acquire the carbon footprint information including first CO.sub.2 emission amount information and second CO.sub.2 emission amount information, the first CO.sub.2 emission amount information being obtained by converting an amount of emission of greenhouse gas concerned with manufacturing the battery pack and the vehicle into a CO.sub.2 emission amount, the second CO.sub.2 emission amount information being obtained by converting an amount of emission of greenhouse gas emitted during primary use of the battery pack in the vehicle into a CO.sub.2 emission amount, and provide a requester with the carbon footprint information in response to a request from the requester.

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Appl. No.: **18/933126**

Filed: **October 31, 2024**

Foreign Application Priority Data

JP	2024-026827	Feb. 26, 2024

Publication Classification

Int. Cl.: **B60L58/10** (20190101); **G06Q30/0283** (20230101)

U.S. Cl.:

CPC **B60L58/10** (20190201); **G06Q30/0283** (20130101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to Japanese Patent Application No. 2024-026827 filed on Feb. 26, 2024, incorporated herein by reference in its entirety.

BACKGROUND

1. Technical Field

[0002] The disclosure relates to an information providing system that provides carbon footprint information of an in-vehicle battery pack.

2. Description of Related Art

[0003] Japanese Unexamined Patent Application Publication No. 2010-254053 (JP 2010-254053 A) describes a greenhouse gas emission display device mounted on a vehicle including different types of driving sources. Japanese Unexamined Patent Application Publication No. 2014-215773 (JP 2014-215773 A) describes a technology to estimate the amount of emission of greenhouse gas emitted from electronic media, such as electronic books, and an information communication device used to browse or reproduce the electronic media. Furthermore, Japanese Unexamined Patent Application Publication No. 2011-123670 (JP 2011-123670 A) describes a greenhouse gas emissions rights trading system.

SUMMARY

[0004] As for battery packs that are estimated to be reused after primary use in vehicles, information on the amount of emission of greenhouse gas emitted during the time from resource mining to primary use is worth for people that are interested in the battery packs, that is, users and other people that reuse battery packs. For this reason, it is important to keep track of the amount of emission and get ready to provide the amount of emission.

[0005] An information providing system according to the disclosure is a system that provides carbon footprint information of a battery pack mounted on a vehicle and includes one or more processors. The one or more processors are configured to acquire the carbon footprint information including first CO.sub.2 emission amount information and second CO.sub.2 emission amount information, the first CO.sub.2 emission amount information being obtained by converting an amount of emission of greenhouse gas concerned with manufacturing the battery pack and the vehicle into a CO.sub.2 emission amount, the second CO.sub.2 emission amount information being obtained by converting an amount of emission of greenhouse gas emitted during primary use of the battery pack in the vehicle into a CO.sub.2 emission amount, and provide a requester with the carbon footprint information in response to a request from the requester.

[0006] With the information providing system according to the disclosure, it is possible to provide information on the amount of emission of greenhouse gas concerned with an in-vehicle battery pack as carbon footprint information.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] Features, advantages, and technical and industrial significance of exemplary embodiments of the disclosure will be described below with reference to the accompanying drawings, in which like signs denote like elements, and wherein:

[0008] FIG. **1** is a conceptual diagram for illustrating a general outline of a battery reuse management system according to an embodiment;
[0009] FIG. **2** is a conceptual diagram for illustrating information related to the reuse of a battery pack;
[0010] FIG. **3** is a diagram that shows processes P1, P2, P3, P4, P5, P6, and P7 related to a battery pack;
[0011] FIG. **4** is a block diagram that shows the functional configuration of a first example related to acquisition and provision of CFP information;
[0012] FIG. **5** is a block diagram that shows the functional configuration of a second example related to acquisition and provision of CFP information; and
[0013] FIG. **6** is a block diagram that shows the functional configuration of a third example related to acquisition and provision of CFP information.

DETAILED DESCRIPTION OF EMBODIMENTS

[0014] An embodiment of the disclosure will be described with reference to the accompanying drawings.

1. Battery Reuse Management System

[0015] FIG. **1** is a conceptual diagram for illustrating a general outline of a battery reuse management system **1** according to the present embodiment. The battery reuse management system **1** manages the reuse of a battery pack **10**. Reuse is a concept including secondary use, tertiary use, ... Particularly, the battery reuse management system **1** manages the reuse of the battery pack (in-vehicle battery pack) **10** mounted on a vehicle **100**. The vehicle **100** is, for example, a battery electric vehicle (BEV) that uses an electric motor as a driving power unit. Alternatively, the vehicle **100** may be, for example, a plug-in hybrid electric vehicle (PHEV) that uses an electric motor and an internal combustion engine as driving power units and that allows the battery pack **10** to be charged from an outside source. The battery pack **10** is mounted on the vehicle **100**. The electric motor is driven by electric power supplied from the battery pack **10**.
[0016] The battery reuse management system **1** includes one or more vehicles **100** and an information management system **300**. The information management system **300** serves as an information platform that manages various pieces of information. The information management system **300** includes one or more servers. The information management system **300** may be made up of a plurality of servers that perform distributed processing. Each vehicle **100** and the information management system **300** can communicate with each other via a wireless communication network.
[0017] More specifically, the information management system **300** includes an information processing device **310**. The information processing device **310** stores various pieces of information, executes various pieces of information processing, and communicates with external sources via a communication interface. The information processing device **310** includes one or more processors and one or more storage devices. The functions of the information processing device **310** may be implemented by cooperation of the processor(s) running a control program with the storage device(s). The control program is stored in the storage device(s). Alternatively, the control program may be recorded on a computer-readable recording medium. The information processing device **310** may be called processing circuitry.
[0018] The vehicle **100** may periodically transmit vehicle information on the vehicle **100** to the information management system **300**. The vehicle information may include travel history information TRV indicating the travel history of the vehicle **100**. For example, the travel history information TRV includes a total travel distance of the vehicle **100**. In another example, the travel history information TRV may include location history information of the vehicle **100**. The location of the vehicle **100** may be expressed by a combination of latitude and longitude.
[0019] The vehicle information may include battery information BAT reflecting a usage history of the battery pack **10**. For example, the battery information BAT includes battery usage history

information. The battery usage history information includes at least one of a temperature history, a state-of-charge (SOC) history, a current history, a voltage history, a charge history, and the like. Each history may be expressed by a frequency distribution. The battery information BAT may include a state of health (SOH). The SOH may be calculated based on battery usage history information. The battery information BAT may include a total charge electricity amount and/or a total discharge electricity amount or may include a total amount of charging power and/or a total amount of discharging power. The battery information BAT may include a battery capacity and/or a battery output.
[0020] The battery reuse management system **1** may further include a terminal **200** that is used at a vehicle dealer or a vehicle repair factory. At a vehicle dealer or a vehicle repair factory, the above-described vehicle information can be read from the vehicle **100** with a predetermined tool. The vehicle information read from the vehicle **100** is input to the terminal **200**. In other words, the terminal **200** can also acquire the vehicle information. The terminal **200** is communicable with the information management system **300** via a wireless or wired communication network. The terminal **200** may transmit vehicle information read from the vehicle **100** to the information management system **300**.
[0021] The information management system **300** collects, accumulates, and manages vehicle information on a large number of vehicles **100**. The information management system **300** includes a vehicle information database **320** that accumulates vehicle information on a large number of vehicles **100**. The vehicle information database **320** is implemented by one or more storage devices.
[0022] Next, the reuse of the battery packs **10** that have been mounted on the vehicles **100** will be described. The battery packs **10** are taken out from the vehicles **100** and reused.
[0023] A battery use system **500** is a system that uses a storage battery. For example, the battery use system **500** is a stationary storage battery system that accumulates renewable energy in a stationary storage battery. In another example, the battery use system **500** may be a home power supply system using a stationary storage battery. According to the present embodiment, the battery packs **10** that have been mounted on the vehicles **100** are reused in such a battery use system **500**. A large number of battery packs **10** are provided from a large number of vehicles **100**, so cost reduction of the battery use system **500** is facilitated, and proliferation of the battery use system **500** is facilitated. Then, the use of renewable energy is facilitated, so CO.sub.2 emission is also reduced.
[0024] FIG. **2** is a conceptual diagram for illustrating information related to the reuse of a battery pack **10**. The battery pack **10** includes one or more battery cells **11** and a controller **12** for controlling the battery pack **10**. The controller **12** is also called electronic control unit (ECU).
[0025] In reusing the battery pack **10**, the battery reuse management system **1** acquires primary use information PU. The primary use information PU corresponds to vehicle information when the battery pack **10** is primarily used in the vehicle **100**.
[0026] For example, the primary use information PU includes battery information BAT reflecting the usage history of the battery pack **10** primarily used in the vehicle **100**. The battery information BAT is stored in the controller **12** (ECU) in the battery pack **10** and is read from the controller **12**. For example, just before the battery pack **10** is taken out from the vehicle **100**, an information processing device of the vehicle **100** acquires the battery information BAT from the controller **12** in the battery pack **10** and transmits the battery information BAT to the information management system **300**.
[0027] The primary use information PU may include travel history information TRV of the vehicle **100** on which the battery pack **10** has been mounted. The travel history information TRV indicates at least one of the total travel distance and location history information of the vehicle **100**. The total travel distance is stored in the controller **12** in the battery pack **10** or a storage device of an in-vehicle system. The location history information is obtained from the storage device of the in-vehicle system or the vehicle information database **320** of the information management system

300. The information management system **300** acquires the travel history information TRV with a method similar to the method of acquiring the battery information BAT.

[0028] Battery management information **30** is information that is used when the battery pack **10** is reused. The battery management information **30** is generated based on the primary use information PU. More specifically, the battery management information **30** includes the battery information BAT and the travel history information TRV. There are various environments (ambient temperature, humidity, and the like) of regions where the vehicle **100** has travelled, and the degradation condition of battery performance varies depending on an environment, so the travel history information TRV can also be useful information when the battery pack **10** is reused. The battery management information **30** may include the vehicle type of the vehicle **100** on which the battery pack **10** has been mounted. The battery management information **30** may be generated in any one of the vehicle **100**, the terminal **200**, and the information management system **300**.

[0029] The information management system **300** collects, accumulates, and manages pieces of battery management information **30** on a large number of vehicles **100**. The information management system **300** includes a battery management information database **330** that accumulates the pieces of battery management information **30** on a large number of vehicles **100** (see FIG. **1**). The battery management information database **330** is implemented by one or more storage devices.

[0030] Reuse identification information (hereinafter, referred to as “reuse ID”) is identification information of the battery pack **10** reused. The reuse ID is associated with the battery pack **10**. The information processing device of the vehicle **100** associates the reuse ID with the primary use information PU or the battery management information **30** and transmits the reuse ID and the primary use information PU or the battery management information **30** to the information management system **300**. In this way, the information management system **300** acquires the reuse ID and the battery management information of the battery pack **10**. The information management system **300** associates the reuse ID with the battery management information **30** in the battery management information database **330**. In other words, the information management system **300** manages the reuse ID and the battery management information **30** in association with each other.

[0031] The battery pack **10** taken out from the vehicle **100** is provided to a user that reuses the battery pack **10** (hereinafter, referred to as “reuse user” or “reuser”). The reuse user is an operator of the battery use system **500**. The information management system **300** provides the reuse ID and the battery management information **30** to the reuse user in association with each other.

[0032] More specifically, the battery use system **500** includes an information collecting apparatus **510**. The information collecting apparatus **510** is communicable with the information management system **300** via a wired or wireless communication network. The information management system **300** transmits the reuse ID and the battery management information **30** of the battery pack **10** to the information collecting apparatus **510** of the battery use system **500** in which the battery pack **10** is reused. The information collecting apparatus **510** manages the reuse ID and the battery management information **30** in association with each other. On the other hand, the reuse ID of the battery pack **10** is stored in the controller **12** in the battery pack **10**. Therefore, in the battery use system **500**, the battery pack **10**, the reuse ID, and the battery management information **30** are associated with one another.

[0033] When the battery pack **10** is reused in the battery use system **500**, the information collecting apparatus **510** monitors the status of use of the battery pack **10** and adds battery usage history information on the battery pack **10** to the battery management information **30** (battery information BAT). As a result, the battery management information **30** includes not only the battery information BAT when the battery pack **10** is primarily used in the vehicle **100** but also the battery information BAT when the battery pack **10** is reused in the battery use system **500**.

[0034] The information collecting apparatus **510** periodically uploads the latest battery management information **30** to the information management system **300** together with the reuse ID. The information management system **300** periodically acquires the latest battery management information **30** and the reuse ID. The information management system **300** updates the battery management information **30** associated with the reuse ID in the battery management information database **330** with the latest one.

[0035] A user that desires to reuse a battery pack **10** is referred to as "reuse applicant". A reuse applicant can be a reuse user in the future. The information management system **300** may provide the reuse ID and the battery management information **30** of a battery pack **10** to a reuse applicant. More specifically, a reuse applicant makes a request of the information management system **300** to provide information with the use of a terminal **400** (see FIG. **1**). The information management system **300** provides the terminal **400** with pieces of battery management information **30** and reuse IDs of various battery packs **10**. The reuse applicant is able to select a desired battery pack **10** (reuse ID) by viewing the pieces of battery management information **30** displayed on the terminal **400**.

2. Provision of CFP Information

[0036] FIG. **3** is a diagram that shows processes P1, P2, P3, P4, P5, P6, and P7 related to a battery pack **10**. The process P1 relates to mining of necessary resources to manufacture battery cells **11** included in a battery pack **10** and manufacturing of necessary materials for manufacturing the battery cells **11**. The process P2 relates to manufacturing of the battery cells **11**. The process P3 relates to manufacturing of the battery pack **10**. Broadly speaking, the processes P1, P2, and P3 relate to manufacturing of the battery pack **10**. The process P4 relates to manufacturing of a vehicle **100** on which the battery pack **10** is mounted. The process P5 relates to primary use of the battery pack **10** in the vehicle **100**. The process P6 relates to the reuse of the battery pack **10** after the end of primary use. Reuse at least includes secondary use and can be further repeatedly used in some cases. The disposal process P7 relates to, for example, collection of the discarded battery pack **10** and various handlings of the collected battery pack **10**. Examples of the various handlings include making the battery cells **11** harmless, and recycling to take out reusable materials from the battery cells **11**. Depending on the battery pack **10**, the reuse process P6 can be omitted.

[0037] In the above-described processes P1, P2, P3, P4, P5, P6, and P7, greenhouse gas, such as CO.sub.2, is emitted or can be emitted. In the present embodiment, the information management system **300** corresponds to an example of the "information providing system" according to the disclosure. Therefore, in order to provide carbon footprint (CFP) information of the battery pack **10**, the information management system **300** acquires (collects) a CO.sub.2 emission amount converted from the amount of emission of greenhouse gas in each of the processes P1, P2, P3, P4, P5, P6, and P7 as CFP information. Then, the information management system **300** provides a requester with the CFP information in response to a request from the requester.

2-1. First Example

[0038] FIG. **4** is a block diagram that shows the functional configuration of a first example related to acquisition and provision of CFP information. The information management system **300** includes a first emission amount acquisition unit **340**, a second emission amount acquisition unit **350**, and a CFP information processing unit **360** that are implemented by the information processing device **310**. CFP information acquired in the first example includes first CO.sub.2 emission amount information and second CO.sub.2 emission amount information.

First Emission Amount Acquisition Unit

[0039] The first emission amount acquisition unit **340** acquires first CO.sub.2 emission amount information and stores the acquired first CO.sub.2 emission amount information in the storage device of the information processing device **310**. The first CO.sub.2 emission amount information is obtained by converting the amount of emission of greenhouse gas concerned with manufacturing of the battery pack **10** and the vehicle **100** into a CO.sub.2 emission amount. The first CO.sub.2 emission amount information includes, for example, the following CO.sub.2 emission amounts A, B, C, and D. The CO.sub.2 emission amount A is a total value of CO.sub.2 emission amounts respectively emitted during resource mining and material manufacturing in the process P1. The

CO_2 emission amount A may be, for example, calculated as a CO_2 emission amount per 1 kg of the material of the battery cell **11**. The CO_2 emission amount B is a CO_2 emission amount emitted during manufacturing of the battery cell **11** in the process P2 and may be, for example, calculated as a CO_2 emission amount required to manufacture a battery cell **11**. Alternatively, the CO_2 emission amount B may be calculated as a CO_2 emission amount required to manufacture a battery stack made up of a plurality of battery cells **11**. The CO_2 emission amount C is a CO_2 emission amount emitted during manufacturing of the battery pack **10** in the process P3. The CO_2 emission amount D is a CO_2 emission amount emitted during manufacturing of the vehicle **100** in the process P4.

[0040] For example, the CO_2 emission amount A can be calculated based on a CO_2 emission amount during production of greenhouse gas and electric power used for resource mining and material manufacturing. Similarly, the CO_2 emission amount B can be calculated based on a CO_2 emission amount during production of greenhouse gas and electric power used to manufacture the battery cell **11**. The CO_2 emission amount C can be calculated based on a CO_2 emission amount during production of greenhouse gas and electric power used to manufacture the battery pack **10**. The CO_2 emission amount D can be calculated based on a CO_2 emission amount during production of greenhouse gas and electric power used to manufacture the vehicle **100**. Additionally, each of the CO_2 emission amounts A, B, C, and D may include a CO_2 emission amount during transport of a corresponding one of the materials, the battery cell **11**, the battery pack **10**, and the vehicle **100** to a place where the next process is performed.

[0041] The above-described CO_2 emission amounts A, B, C, and D are calculated by, for example, business operators that are respectively in charge of the processes P1, P2, P3, and P4. Some of the business operators concerned with at least two processes of the process P1, the process P2, the process P3, and the process P4 may be the same (for example, a manufacturing business operator for the battery pack **10** and a manufacturing business operator for the vehicle **100**). Then, as shown in FIG. **4**, the first emission amount acquisition unit **340** communicates with individual terminals **600** of the business operators to acquire CO_2 emission amounts A, B, C, and D (that is, the first CO_2 emission amount information). Alternatively, in an example, in which the manufacturing business operator for the vehicle **100** acquires the upstream CO_2 emission amounts A, B, and C in advance, the first emission amount acquisition unit **340** communicates with the terminal **600** of the manufacturing business operator for the vehicle **100** to acquire the CO_2 emission amounts A, B, C, and D. Alternatively, the first emission amount acquisition unit **340** may acquire information that is a basis for calculating the CO_2 emission amounts A, B, C, and D from the business operators and calculate the CO_2 emission amounts A, B, C, and D based on the information in itself.

Second Emission Amount Acquisition Unit

[0042] The second emission amount acquisition unit **350** acquires second CO_2 emission amount information and stores the acquired second CO_2 emission amount information in the storage device of the information processing device **310**. The second CO_2 emission amount information is obtained by converting the amount of emission of greenhouse gas emitted during primary use (more specifically, during the entire period of primary use) of the battery pack **10** in the vehicle **100** into a CO_2 emission amount. Here, a method of calculating second CO_2 emission amount information will be described by way of a BEV and a PHEV as examples sequentially.

[0043] Initially, in an example of a BEV, a CO_2 emission amount E during production of electric power for charging the battery pack **10** with an external charger corresponds to the second CO_2 emission amount information. The CO_2 emission amount E is a product of a total amount of charging power (kWh) of the battery pack **10** during primary use and a CO_2 emission coefficient Ke. The CO_2 emission coefficient Ke is, for example, a CO_2

emission coefficient (t-CO_2/kWh) of electric power per unit amount of electric power. The total amount of charging power used to calculate a CO_2 emission amount E can be, for example, acquired as follows. In other words, in a case where the battery information BAT that can be acquired from the vehicle **100** includes a total charge electricity amount (Ah), the total amount of charging power can be acquired by dividing a product of the total charge electricity amount derived from the battery information BAT and a total voltage (V) of a main power supply for power by 1000. The total voltage of the main power supply for power may be, for example, a value released by the manufacturing business operator for the vehicle **100** (so-called catalog value). In a case where the battery information BAT includes a total amount of charging power itself, the total amount of charging power can be directly acquired from the battery information BAT. For example, a numeric value released every year by an external institution for each electric power supplier may be used as the CO_2 emission coefficient Ke. The storage device of the information processing device **310** stores data of the CO_2 emission coefficient Ke for each electric power supplier, acquired through communication with a server of an external institution.
[0044] The second emission amount acquisition unit **350**, for example, calculates a product of one-year amount of charging power and the CO_2 emission coefficient Ke as the CO_2 emission amount E every year during primary use of the battery pack **10**. Then, the second emission amount acquisition unit **350** adds up the CO_2 emission amount E of each year and calculates the CO_2 emission amount E that is a total during primary use as the second CO_2 emission amount information. Additionally, the CO_2 emission coefficient Ke decreases as the percentage of the amount of renewable energy generated (the percentage of renewable energy) in the amount of electric power generated by an electric power supplier that supplies electric power to an external charger increases. For example, when the percentage is 100%, the CO_2 emission coefficient Ke is zero, so the CO_2 emission amount E is also zero. For this reason, the CO_2 emission amount E may be calculated by using the CO_2 emission coefficient Ke that reflects the percentage. An electric power supplier that supplies electric power to an external charger can be identified based on the location information of the vehicle **100** being externally charged.
[0045] Alternatively, the second emission amount acquisition unit **350** may acquire the CO_2 emission amount E not every year and may acquire the CO_2 emission amount E each time the battery pack **10** is charged. In other words, at the end of charging, the second emission amount acquisition unit **350** may acquire information on a charge electricity amount and location information from the vehicle **100** and calculate a current value of the CO_2 emission amount E based on the acquired information. Then, the second emission amount acquisition unit **350** may update an accumulated value of the CO_2 emission amount E (that is, a second CO_2 emission amount) by adding the current value to the accumulated value until the last charging. Alternatively, the second emission amount acquisition unit **350** may calculate a CO_2 emission amount E based on information on the total amount of charging power acquired from the vehicle **100** at the end of primary use. The information processing device of the vehicle **100** may calculate the CO_2 emission amount E. In this example, the second emission amount acquisition unit **350** may, for example, communicate with the vehicle **100** at the end of primary use and acquire the CO_2 emission amount E from the vehicle **100**.
[0046] Next, in an example of a PHEV, the sum of the above-described CO_2 emission amount E and the next CO_2 emission amount F corresponds to the second CO_2 emission amount information. A PHEV is capable of driving using only an electric motor without operating an internal combustion engine (BEV driving) and driving by cooperation between the internal combustion engine and the electric motor (HEV driving). The CO_2 emission amount E corresponds to a CO_2 emission amount in BEV driving, and the CO_2 emission amount F corresponds to a CO_2 emission amount in HEV driving. The CO_2 emission amount F is a product of a value obtained by dividing a travel distance DH (km) of the vehicle **100** using HEV

driving by a fuel consumption rate Fe (km/L), and the CO_2 emission amount X per 1 L of fuel. For example, a value included in the travel history information TRV that can be acquired from the vehicle **100** may be used as the travel distance DH. For example, a value released by a manufacturing business operator for the vehicle **100** (so-called catalog value) may be used as the fuel consumption rate Fe. The CO_2 emission amount X can be acquired by, for example, communicating with the server of an external institution. The acquired fuel consumption rate Fe and CO_2 emission amount X are stored in the storage device of the information processing device **310**.

[0047] The second emission amount acquisition unit **350** periodically communicates with the vehicle **100** during primary use to acquire the travel distance DH after last acquisition of data. The second emission amount acquisition unit **350** calculates a current value of the CO_2 emission amount F based on the acquired travel distance DH, fuel consumption rate Fe, and CO_2 emission amount X. Then, the second emission amount acquisition unit **350** updates an accumulated value of the CO_2 emission amount F by adding a current value to the accumulated value at the time of calculation of a last value of the CO_2 emission amount F. Then, the second emission amount acquisition unit **350** periodically calculates the sum of the separately calculated accumulated value of the CO_2 emission amount E and accumulated value of the CO_2 emission amount F and updates the second CO_2 emission amount information with the sum.

[0048] Alternatively, the CO_2 emission amount F may be calculated by using a total travel distance DH acquired from the vehicle **100** at the end of primary use. The CO_2 emission amount F may be calculated by the information processing device of the vehicle **100**. In this example, the second emission amount acquisition unit **350** may, for example, communicate with the vehicle **100** at the end of primary use and acquire the CO_2 emission amount F from the vehicle **100**.

[0049] Additionally, the second CO_2 emission amount information calculated as described above may include a CO_2 emission amount at the time of transporting the battery pack **10** detached from the vehicle **100** at the end of primary use to a place of reuse.

CFP Information Processing Unit

[0050] In the first example, the CFP information processing unit **360** acquires the latest value of the first CO_2 emission amount information from the first emission amount acquisition unit **340** and the latest value of the second CO_2 emission amount information from the second emission amount acquisition unit **350** and manages the acquired latest values as CFP information. The CFP information processing unit **360** may, for example, individually manage the latest value of the first CO_2 emission amount information and the latest value of the second CO_2 emission amount information as CFP information. The CFP information to be managed may include a total value of the latest value of the first CO_2 emission amount information and the latest value of the second CO_2 emission amount information. Alternatively, the CFP information to be managed may be only the total value.

[0051] The information management system **300** is communicable with a terminal **700** that is used by a requester that desires to be provided with CFP information. The terminal **700** is, for example, a personal computer or a smartphone. The CFP information processing unit **360** provides CFP information via the terminal **700** to the requester that operates the terminal **700** in response to a request from the requester. Transmission of the CFP information to the requester or permission of access of the requester to the CFP information using the terminal **700** corresponds to an example of providing CFP information.

[0052] A requester is, for example, a secondary use applicant (reuse applicant) that desires to secondarily use a battery pack **10**. In an example in which the requester is a secondary use applicant, the terminal **700** (see FIG. **4**) is the same as the terminal **400** (see FIG. **1**). Additionally, CFP information may be provided to the secondary use applicant together with the reuse ID and the battery management information **30** of the battery pack **10**.

[0053] As shown in FIG. **4**, the CFP information processing unit **360** may include a price setting unit **361**. The price setting unit **361** sets the price of the battery pack **10** to be presented to a secondary use applicant according to CFP information. Specifically, the price may be, for example, set so as to increase as a numeric value of CFP information (a numeric value of at least one of the first CO_2 emission amount information and the second CO_2 emission amount information) decreases. More specifically, the second CO_2 emission amount information used to set the price covers the entire period of primary use. In an example of a BEV in which the CO_2 emission amount E corresponds to the second CO_2 emission amount information, the price may be set so as to increase as a value obtained by dividing the CO_2 emission amount E by the total amount of charging power of the battery pack **10** during primary use (that is, the CO_2 emission coefficient Ke) decreases. Similarly, in an example of a PHEV in which the sum of the CO_2 emission amount E and the CO_2 emission amount F correspond to the second CO_2 emission amount information as well, the price may be set in consideration of the fact that the price is increased as the CO_2 emission coefficient Ke decreases.

[0054] Alternatively, the requester may be, for example, a secondary use user (reuse user). The CFP information processing unit **360** may, for example, provide CFP information to a secondary use user in response to a request in advance after the secondary use user is determined as a destination to provide a battery pack **10**. Alternatively, provision of CFP information may be performed in response to a request after the start of secondary use.

[0055] Alternatively, the requester may be, for example, a consumer. Here, a consumer is, for example, a user that owns another vehicle **100** of the same vehicle type as the vehicle **100** on which the battery pack **10** having CFP information to be provided, or a user that desires to purchase the another vehicle **100**.

[0056] Additionally, provision of CFP information to a requester may be, for example, free or paid. Whether provision is free or paid may vary depending on a requester. In an example in which CFP information is periodically updated during primary use of the battery pack **10**, the CFP information processing unit **360** may receive an application from a requester during primary use of the battery pack **10**. Then, the CFP information processing unit **360** may provide a requester with CFP information including the latest second CO_2 emission amount at the timing of application by the requester.

Advantageous Effects

[0057] According to the first example related to acquisition and provision of CFP information, the first CO_2 emission amount information during the period from resource mining to manufacturing of the vehicle **100** and the second CO_2 emission amount information during primary use can be provided to a requester as CFP information in response to a request.

[0058] Receiving provision of CFP information until primary use has the following advantage for a secondary use user and a secondary use applicant of the battery pack **10**. In other words, in a case where CFP information until primary use is excellent, a business operator that secondarily uses the battery pack **10** can clearly call attention to society in constructing a low-carbon storage battery system using the battery pack **10** with a small CO_2 emission amount. In an example in which a manufacturing business operator for the vehicle **100** is a business operator that manages the information management system **300** (information providing system), providing CFP information in response to a request has the following advantage for the manufacturing business operator. In other words, in a case where CFP information is excellent, the manufacturing business operator can clearly call attention to society in that the low-carbon battery pack **10** is mounted on the vehicle **100** manufactured by itself.

[0059] With the price setting unit **361**, the price of the battery pack **10** to be present to a secondary use applicant is set according to CFP information. For a business operator that secondarily uses the battery pack **10**, excellent CFP information has the above-described advantage. Therefore, the battery pack **10** having excellent CFP information has a higher value than the battery pack **10** having not excellent CFP information. For this reason, by setting a price according to CFP information, dealing of the battery pack **10** at a further appropriate price reflecting a value based on the viewpoint of CFP information is facilitated.

2-2. Second Example

[0060] FIG. **5** is a block diagram that shows the functional configuration of a second example related to acquisition and provision of CFP information. The second example differs from the first example in that the information management system **300** additionally includes a third emission amount acquisition unit **370**.

[0061] The third emission amount acquisition unit **370** acquires third CO.sub.2 emission amount information and stores the acquired third CO.sub.2 emission amount information in the storage device of the information processing device **310**. The third CO.sub.2 emission amount information is obtained by converting the amount of emission of greenhouse gas emitted during reuse of the battery pack **10** into a CO.sub.2 emission amount.

[0062] Specifically, a CO.sub.2 emission amount G during production of electric power for charging the battery pack **10** during reuse of the battery pack **10** corresponds to the third CO.sub.2 emission amount information. A method of calculating the CO.sub.2 emission amount G is, for example, similar to the above-described method of calculating the CO.sub.2 emission amount E. In other words, the CO.sub.2 emission amount G is, for example, a product of a total amount of charging power (kWh) of the battery pack **10** during reuse and a CO.sub.2 emission coefficient Ke. The total amount of charging power (kWh) can be, for example, acquired from the battery management information **30** of the battery pack **10**, periodically transmitted from the information collecting apparatus **510** of the battery use system **500**. More specifically, even when the total amount of charging power is the same, the CO.sub.2 emission amount G reduces as the above-described renewable energy percentage increases. The CO.sub.2 emission amount G may be calculated by the information processing device of the battery use system **500**. In this example, the third emission amount acquisition unit **370** may, for example, communicate with the battery use system **500** at the end of secondary use and acquire the CO.sub.2 emission amount G from the battery use system **500**.

[0063] Additionally, the third CO.sub.2 emission amount information calculated as described above may include a CO.sub.2 emission amount at the time of transporting the battery pack **10** from a place of secondary use to a place of tertiary use or disposal.

[0064] In the second example, the CFP information processing unit **360** acquires the latest value of the first CO.sub.2 emission amount information from the first emission amount acquisition unit **340**, the latest value of the second CO.sub.2 emission amount information from the second emission amount acquisition unit **350**, and the latest value of the third CO.sub.2 emission amount information from the third emission amount acquisition unit **370** and manages the acquired latest values as CFP information. The CFP information processing unit **360** may, for example, individually manage the latest value of the first CO.sub.2 emission amount information, the latest value of the second CO.sub.2 emission amount information, and the latest value of the third CO.sub.2 emission amount information as CFP information. The CFP information to be managed may include a total value of the latest value of the first CO.sub.2 emission amount information, the latest value of the second CO.sub.2 emission amount information, and the latest value of the third CO.sub.2 emission amount information. Alternatively, the CFP information to be managed may be only the total value.

[0065] In an example in which tertiary use is performed after secondary use, the third emission amount acquisition unit **370**, as in the case of the CO.sub.2 emission amount G during secondary use, calculates a CO.sub.2 emission amount H during tertiary use and adds the calculated CO.sub.2 emission amount H to the third CO.sub.2 emission amount information. This also applies to a case where reuse, such as quaternary use, is performed.

[0066] As for provision of CFP information until primary use, a requester in the second example is, for example, the same as the requester described for the first example. As for provision of only CFP information during secondary use (third CO_2 emission amount information) or total CFP information until secondary use, a requester may include the following person. In other words, in a case where tertiary use is performed after secondary use of the battery pack **10**, a requester in the second example may be a secondary use applicant (reuse applicant) that desires tertiary use. As in the case of the example of the secondary use applicant, the price setting unit **361** may set the price of the battery pack **10** to be provided for a tertiary use applicant according to CFP information. Then, the requester may be, for example, a tertiary use user (reuse user). These also apply to a case where reuse, such as quaternary use, is performed.
[0067] According to the second example described above, it is possible to provide a requester with CFP information including the third CO_2 emission amount information together with the first CO_2 emission amount information and the second CO_2 emission amount information in response to a request.
[0068] As in the case of the example of secondary use, a business operator that tertiarily uses the battery pack **10** can also clearly call attention to society by receiving provision of CFP information until primary use or secondary use in constructing a low-carbon storage battery system. This also applies to a case where reuse, such as quaternary use, is performed.
2-3. Third Example
[0069] FIG. **6** is a block diagram that shows the functional configuration of a third example related to acquisition and provision of CFP information. The third example differs from the second example in that the information management system **300** additionally includes a fourth emission amount acquisition unit **380**.
[0070] The fourth emission amount acquisition unit **380** acquires fourth CO_2 emission amount information and stores the acquired fourth CO_2 emission amount information in the storage device of the information processing device **310**. The fourth CO_2 emission amount information is obtained by converting the amount of emission of greenhouse gas emitted during the disposal process P7 of the battery pack **10** into a CO_2 emission amount.
[0071] Specifically, a CO_2 emission amount I that is a total value of the CO_2 emission amounts respectively emitted at the time of collecting the battery pack **10** and the various handlings (such as recycling of the materials of the battery cells **11**), included in the disposal process P7, corresponds to the fourth CO_2 emission amount information. For example, the CO_2 emission amount I can be calculated based on a CO_2 emission amount during production of greenhouse gas and electric power used for collection of the battery pack **10** and the various handlings. Additionally, the CO_2 emission amount I may include a CO_2 emission amount at the time of transporting recycled materials to a place where the next process (that is, the process P2 of manufacturing the battery cell **11** using the recycled materials) is performed.
[0072] The information management system **300** is communicable with one or more terminals **800** used by one or more business operators related to the disposal process P7 (see FIG. **3**) of the battery pack **10**. The fourth emission amount acquisition unit **380** communicates with the one or more terminals **800** to acquire the CO_2 emission amount I. Alternatively, the fourth emission amount acquisition unit **380** may acquire information that is a basis for calculating the CO_2 emission amount I from the one or more business operators and calculate the CO_2 emission amount I based on the information in itself.
[0073] In the third example, the CFP information processing unit **360** acquires the latest value of the first CO_2 emission amount information from the first emission amount acquisition unit **340**, the latest value of the second CO_2 emission amount information from the second emission amount acquisition unit **350**, the latest value of the third CO_2 emission amount information from the third emission amount acquisition unit **370**, and the latest value of the fourth CO_2 emission amount information from the fourth emission amount acquisition unit **380** and

manages the acquired latest values as CFP information. The CFP information processing unit **360** may, for example, individually manage the latest value of the first CO.sub.2 emission amount information, the latest value of the second CO.sub.2 emission amount information, the latest value of the third CO.sub.2 emission amount information, and the latest value of the fourth CO.sub.2 emission amount information as CFP information. The CFP information to be managed may include a total value of the latest value of the first CO.sub.2 emission amount information, the latest value of the second CO.sub.2 emission amount information, the latest value of the third CO.sub.2 emission amount information, and the latest value of the fourth CO.sub.2 emission amount information. Alternatively, the CFP information to be managed may be only the total value.
[0074] As for provision of CFP information until reuse, such as secondary use, a requester in the third example is, for example, the same as the requester described for the first and second examples. As for provision of CFP information of all the processes from the processes P1, P2, P3, P4, P5, P6, and P7, a requester may be, for example, a manufacturing business operator for the vehicle **100** in a case where a business operator different from the manufacturing business operator for the vehicle **100** manages the information management system **300** (information providing system). As for provision of only CFP information of the disposal process P7 (fourth CO.sub.2 emission amount information), a requester may be, for example, a reuse user having reused the battery pack **10** proceeding to the disposal process P7.
[0075] According to the third example described above, it is possible to provide a requester with CFP information including the fourth CO.sub.2 emission amount information together with the first CO.sub.2 emission amount information, the second CO.sub.2 emission amount information, and the third CO.sub.2 emission amount information in response to a request.
[0076] In an example in which a manufacturing business operator for the vehicle **100** is a business operator that manages the information management system **300** (information providing system), collecting the total of the first CO.sub.2 emission amount information to the fourth CO.sub.2 emission amount information from resource mining to disposal as CFP information in a state being ready for provision has the following advantage. In other words, a manufacturing business operator for the vehicle **100** can clearly call attention in addressing low-carbonization by itself and social contribution. In an example in which a manufacturing business operator for the vehicle **100** is different from a business operator that manages the information management system **300**, providing total CFP information from resource mining to disposal by the information management system **300** has the following advantage for the manufacturing business operator. In other words, a manufacturing business operator for the vehicle **100** is able to acquire insufficient CO.sub.2 emission amount information (for example, the fourth CO.sub.2 emission amount information) from the information management system **300** and keep track of total CFP information from resource mining to disposal. Then, the manufacturing business operator can clearly call attention by using the CFP information in addressing low-carbonization by itself and social contribution.

Claims

1. An information providing system that provides carbon footprint information of a battery pack mounted on a vehicle, the information providing system comprising one or more processors, the one or more processors being configured to: acquire the carbon footprint information including first CO.sub.2 emission amount information and second CO.sub.2 emission amount information, the first CO.sub.2 emission amount information being obtained by converting an amount of emission of greenhouse gas concerned with manufacturing the battery pack and the vehicle into a CO.sub.2 emission amount, the second CO.sub.2 emission amount information being obtained by converting an amount of emission of greenhouse gas emitted during primary use of the battery pack in the vehicle into a CO.sub.2 emission amount; and provide a requester with the carbon footprint information in response to a request from the requester.

2. The information providing system according to claim 1, wherein the carbon footprint information further includes third CO_2 emission amount information obtained by converting an amount of emission of greenhouse gas emitted during reuse of the battery pack into a CO_2 emission amount.
3. The information providing system according to claim 1, wherein the requester includes a user that reuses the battery pack or a user that desires to reuse the battery pack.
4. The information providing system according to claim 1, wherein: the requester is a user that desires to reuse the battery pack; and the one or more processors are configured to set a price of the battery pack to be presented to the user in response to the carbon footprint information.
5. The information providing system according to claim 1, wherein the carbon footprint information further includes fourth CO_2 emission amount information obtained by converting an amount of emission of greenhouse gas emitted in a disposal process of the battery pack into a CO_2 emission amount.